



COMPUTING FOR VOLUNTARY WELFARE ORGANISATIONS

IT SYSTEMS FOR SINGAPORE VWOS, CODED WITH LOVE

CODER-POWER

Moderators



Eugene

CVWO '25



Yihao

CVWO '25



Oad

CVWO '25



Florian

CVWO '25



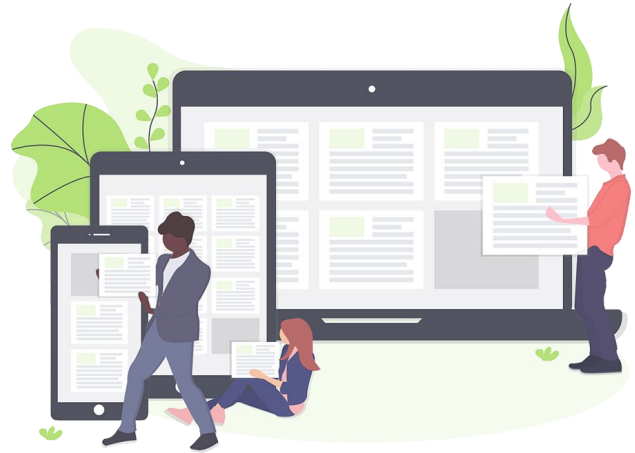
Sai

CVWO '25

Agenda

- What is CVWO?
- Past Year Projects
- Expectations
- Why CVWO
- CVWO Application
- Q&A

What is CVWO?



What is CVWO?

- Computing for **V**oluntary **W**elfare **O**rganisations
- 3 months full-time **summer internship** programme
- Interns work in teams to **develop applications for VWOs**
- Majority of interns are **freshmen** from SoC

What is CVWO?

- Initiated in 2007 by **A/Prof Ben Leong**
- Applications developed by CVWO have been used by **more than 10 beneficiaries** over **more than 30 centres** in Singapore
- Director, AI Centre for Educational Technologies (AICET)



A/Prof Ben Leong

“Our mission is to build IT systems that help our partner VWOs serve the community more effectively.



Past Year Projects



Active Ageing Centres

AAC/SAC



- Active Ageing Centres / Senior Activity Centres are eldercare centres which act as communal spaces for the elderly to **socialise with their peers**, and find **emotional and psychological support**
- CVWO's system allows staff members to easily track the centre's activities and the people they serve, as well as generating reports for audits

Our Journey with AACs

- The first **CVWO system** was built for **AAC** back in 2007.
- We are **AIC-whitelisted** and continuously upgrading our systems to meet new requirements.



Project #1 – AIC v1.2

- Integration with **AIC Centralised API Gateway** for a more efficient workflow
- **Automated retrieval** of:
 - Service boundaries
 - Senior cohorts
 - Incoming referrals
- **Automated submission** of AAC reporting data

Automated Data Pipeline

Testing Client X

VIEW CLIENT

Import Status

✔ Client already exists and has no conflicts.

Demographic Details

NRIC G1357924K	Name Testing Client X	Date of Birth 1941-05-07	Gender Male
Nationality Singaporean	Race Chinese	Citizenship Status Permanent Resident	Address 146 Braddell Road, #24-176 The Medallion, Singapore 359959

PHV Details

Senior LIN 23882	Senior GRC TANJONG PAGAR GRC	Senior Division TANJONG PAGAR-TIONG BAHRU	Languages Spoken / Understood English, Mandarin, Hokkien, Hakka
Occupation Status Full-time Salaried	Contacts 94558893	NOK Salutation Mrs	NOK Name NOK TWO
NOK Contact 93300920	NOKs —	Preferred Days to Call or Visit Weekday	Preferred Times to Call or Visit Morning
Cognitive Skills Need help. Modified independence or any impairment	Instability of Conditions Yes	Dyspnoea Present at rest	How Client Rates Health Fair

Home / Age Well SG Report Submissions / Submission 2

Submission Details

DOWNLOAD REPORT

64 6 8 82

Submission Status

Search... SEARCH

Report

Client Elderly With Consent 5	Pending Retry
Client Elderly With Consent 6	Pending Retry
Person Client Non-Elderly 13	Pending Retry
Person Client Elderly 19	Pending Retry
Person Client Non-Elderly 14	Pending Retry
Person Client Non-Elderly 12	Pending Retry
Client Elderly With Consent 2	Failed
Person Client Non-Elderly 15	Failed
Client Non-Elderly With Consent 12	Failed



Select a report from the left

Customisable By Clients

Integration Settings

Senior Cohort Listing Automated Import Settings

Last Import Date: There is no last import date

Next Import Date: Fri, 1 Aug 2025, 2:00 AM

Senior Cohort Import Interval
Every Month

Senior Cohort Import Start Date
01/01/2025

IMPORT RECORDS

Service Boundary Automated Import Settings

Last Import Date: There is no last import date

Next Import Date: Fri, 1 Aug 2025, 3:00 AM

Service Boundary Import Interval
Every Month

Service Boundary Import Start Date
01/01/2025

IMPORT SERVICE BOUNDARIES

ASG Reports Automated Submission Settings

Monthly ASG Reports for the reporting period of the previous month will be generated and pending submission every month from the start date.

Quarterly ASG Reports for the reporting period of the previous quarter will be generated and pending submission every 3 months from the start date

Last Automatic Monthly Generation Date: There is no last monthly submission

Next Automatic Monthly Generation Date: Fri, 1 Aug 2025, 4:00 AM

Last Automatic Quarterly Generation Date: There is no last quarterly generation

Next Automatic Quarterly Generation Date: Wed, 1 Oct 2025, 4:00 AM

ASG Reports Generation Start Date
01/01/2025

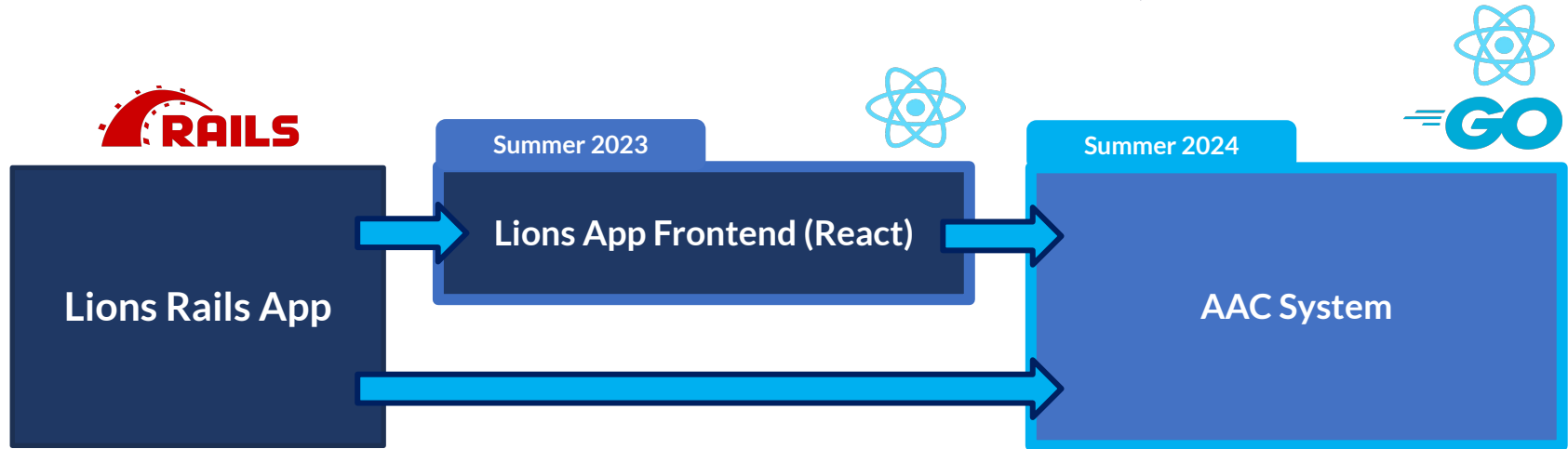
GENERATE REPORTS

REVERT CHANGES

SAVE CHANGES

Project #2 – HPC Porting

- A **multi-year project** completing in **Summer 2025**, with the **H**ome **P**ersonal **C**are module successfully migrated from Rails to the main AAC React and Go system.



Service Sessions

Service Records

[+ NEW SERVICE RECORD](#)

Advanced Search

Session date between and

Status

[EXPORT TO EXCEL](#)[ADVANCED SEARCH](#)

If no date range is specified, only service records for the current month will be shown.

[SHOW/HIDE COLUMNS](#)

Search by client name or client reference number:

S/N	Client Name	Case Number	Status	Service Date	Service Start	Service End	Duration (hours)	Remarks	Staff Name	Actions
1			Service Provided	Tue, 01 Jul 2025	04:14 PM	05:14 PM	1:00	(BP: 120/80, Pulse: 46, Temp: 123) Test	CVWO Admin	>
2			Service Provided	Tue, 01 Jul 2025	04:11 PM	05:11 PM	1:00	(BP: 120/80, Pulse: 36.5, Temp: 80) Test	CVWO Admin	>
3			Service Provided	Tue, 01 Jul 2025	04:11 PM	05:11 PM	1:00	(BP: 120/80, Pulse: 80, Temp: 36.5) Test	CVWO Admin	>
4			Service Provided	Tue, 01 Jul 2025	04:09 PM	05:09 PM	1:00	(BP: 120/80, Pulse: 70, Temp: 36) Test	CVWO Admin	>
5			Service Provided	Tue, 01 Jul 2025	04:03 PM	05:03 PM	1:00	(BP: 120/80, Pulse: 80, Temp: 36.5) Test	CVWO Admin	>
6			Service Provided	Tue, 01 Jul 2025	04:02 PM	05:02 PM	1:00	(BP: 120/80, Pulse: 80, Temp: 36.5) Test	CVWO Admin	>
7			Service Provided	Tue, 01 Jul 2025	04:01 PM	05:01 PM	1:00	(BP: 120/80, Pulse: 80, Temp: 36) Test	CVWO Admin	>
8			Service Provided	Tue, 01 Jul 2025	03:55 PM	04:55 PM	1:00	(BP: 120/80, Pulse: 70, Temp: 36) Test	CVWO Admin	>
9			Voided	Tue, 01 Jul 2025	01:48 PM	02:48 PM	1:00	(BP: 120/80, Pulse: 70, Temp: 36) Test	CVWO Admin	>



Service Record

Client's Name:

Client's Case No: 59

HCA: CVWO Admin

Date: 25/07/2025

Time Session Started: 17:16

Time Session Ended: 18:16

Duration of Session: 1 hour

Services Provided: ADL/iADL, Elder Sitting/Housekeeping

Remarks: (BP: 120/80, Pulse: 70, Temp: 36.5) Test

[Print](#)[Email](#)

Justification for Requesting Void: *

[Request Void](#)

Accounting Module

Home / HPC / Review Routine Statements

1 new transaction posted after statement period

Posted Date	Service Date	Description	Amount	Payment	Balance
02/06/25	21/05/25	Bank Transfer Payment (Receipt No.)		\$ 18.00	\$ 0.00

REPLACE CURRENT STATEMENT TO INCLUDE TRANSACTIONS

Statement Period
Thu, 01 May 2025 – Sat, 31 May 2025

Posted Date	Service Date	Description	Amount	Payment	Balance
		Balance B/F			\$ 18.00
07/05/25	07/05/25	Service Session	\$ 6.00		\$ 24.00
14/05/25	14/05/25	Service Session	\$ 6.00		\$ 30.00
22/05/25	21/05/25	Service Session	\$ 6.00		\$ 36.00
28/05/25	28/05/25	Service Session	\$ 6.00		\$ 42.00
		Total Amount Payable			\$ 42.00

Client Outstanding Balance

0 – 30 Days	31 – 60 Days	61 – 90 Days	91 – 120 Days	> 120 Days	Amount Due
\$ 24.00	\$ 18.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 42.00

Navigation

Period: 05/25 Statement Status: All

Statement #8 of 36

DOWNLOAD ALL STATEMENTS

Current Status

✉ Yet to be emailed
✓ Yet to be reviewed

View



Actions

✉ EMAIL TO CLIENT

✓ MARK AS REVIEWED

Statement Date: 24/07/2025
Reference No: 92
Statement Period: 01/04/2025 - 30/04/2025

Date	Description	Amount	Payment	Balance
	Balance Brought Forward			\$30.00
11/04/2025	Bank Transfer Payment (Receipt No.)	Voided \$12.00		\$42.00
01/04/2025	Bank Transfer Payment (Receipt No.)		\$24.00	\$18.00
04/04/2025	Service Session	\$6.00		\$24.00
11/04/2025	Service Session	\$6.00		\$30.00
24/04/2025	Bank Transfer Payment (Receipt No.)		\$6.00	\$24.00
25/04/2025	Service Session	\$6.00		\$30.00
	Total Amount Payable			\$30.00

0-30 days	31-60 days	61-90 days	91-120 days	>120 days	Total Amount
\$18.00	\$12.00	\$0.00	\$0.00	\$0.00	\$30.00

Payments by cheque should be crossed and made payable to:
CWO HPC
and mail it to,
13 Computing Drive
Singapore

For payments via bank transfer:
Bank Account No: 001245678
Kindly notify us with the transaction reference code once
payment has been made



Kwong Wai Shiu Hospital Next-of-Kin App

Kwong Wai Shiu Hospital

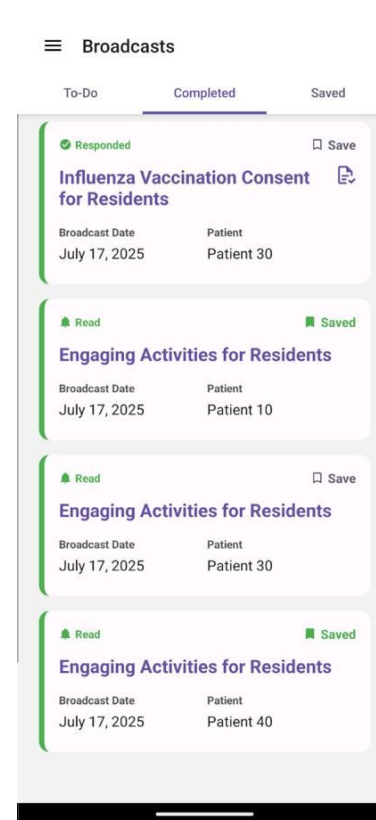
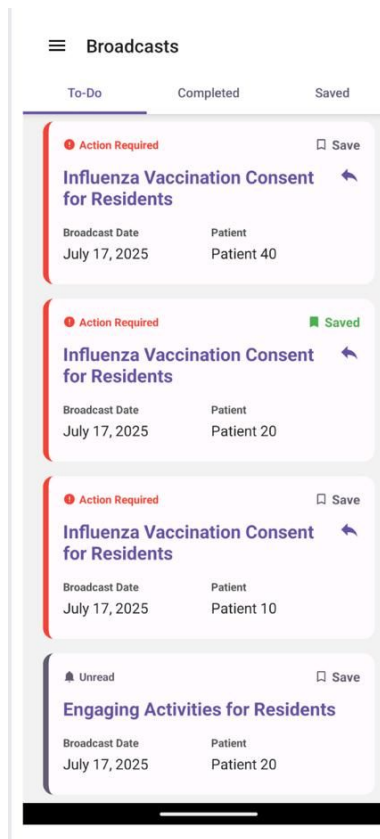
- One of the **oldest charitable healthcare organisations** in Singapore
- **1100 beds** across two nursing homes in Potong Pasir and Serangoon



A Start towards a Super App..

- In summer 2025, KWSH approached CVWO to create their **first client-facing app**.
- Nurses were bogged with having to **call next of kins of patients individually** to seek consent.

The Product



Broadcast management

Broadcast Details

Status History

Template
Vaccination Consent Form

Response Required
Yes

Ward
Ward 9

Selected NOK Roles
Financial Guarantor 1, Financial Guarantor 2, Financial Guarantor 3,
Financial Guarantor 4, Emergency Contact 1, Emergency Contact 2,
Emergency Contact 3, Emergency Contact 4, Main Spokesperson 1, Main
Spokesperson 2

Responses Received
1/10

The broadcast will send another notification to people who haven't completed the form at **18 Jul 2025, 02:24 PM**

RESCHEDULE REMINDER

REMIND NOW 

Responses

SHOW/HIDE COLUMNS

DOWNLOAD EXCEL

S/N	Patient	Responder	Responded At ↓	Influenza Vaccination Consent	View Signature
1	Patient 5	NOK 10	Fri, 18 Jul 2025, 2:25 PM	Consent Given	>
2	Patient 42	—	No Response Yet	—	>

22

Live messaging

 GUY242@GMAIL.COM 

Home / Messaging / Conversations

Active

Closed

 Search conversations...

P3

Patient 3
Next Of Kin 2, Next Of Kin 12, Nex...
We'll continue to monitor closely and ke...
11:47

P1

Patient 13
Next Of Kin 4, Next Of Kin 13
We have something to talk about ...
11:46



Patient 3
Next Of Kin 2, Next Of Kin 12, Next Of Kin 22, Next Of Kin 32

MANAGE

CLOSE

4:07 PM

Yesterday

Today

N

Next Of Kin 2
Thank you for keeping me updated
11:44 AM

P

Hi
11:29 AM

P

Mr Patient 3's condition has stabilised
11:34 AM

P

We will continue monitoring
11:35 AM

P

We'll continue to monitor closely and keep you informed of any changes.
11:47 AM

Type a message...



 **Patient 3 | Ward 4**
Next Of Kin 2, Next Of Kin 12, Next Of Kin 22, Ne...

4:07 PM

Person Staff 1

This is a test message 1
4:07 PM

Today

Person Staff 2

Hi
11:29 AM

Mr Patient 3's condition has stabilised
11:34 AM

We will continue monitoring
11:35 AM

Thank you for keeping me updated
11:44 AM

Person Staff 2

We'll continue to monitor closely and keep you informed of any changes.
11:47 AM

Type your message here...





Counselling Case Management System

Counselling CMS

- CVWO's **Case Management System** helps counselling centres manage clients, cases, and case notes from counselling sessions
- It is a **highly-configurable, multi-tenanted system**
 - It supports multiple different counselling centres, and each centre's website is customised to their specific needs

Counselling CMS

[Home](#) / [Cases](#) / [26/10/1](#) / Case Recording: Unspecified Type

Case Recording Details

Title *

Recording Type *

Session

Session Type(s)

Session Mode *

Date *

02/11/2025



Start Time

7:45 PM

End Time

7:50 PM (5m)

Participants

Select Participant *

Clients

Jerry Tan

Association:

Case

26/10/1



Staff

Person Staff 1

Full-time Staff



ATTENDED

DEFAULTED

Case Notes

Apply Template

TEMPLATE

PAST CASE NOTES

Counselling CMS



FOUNDED IN 1902

Children's Aid Society

Session Type(s)



SELECT ALL

- ☐ Expressive Therapy (Art, Body-based, Play)
- ☐ Counselling
- ☐ Individual
- ☐ Group
- ☐ Family
- ☐ Couple
- ☐ AT
- ☐ PT



Haven Counselling Centre

Session Type(s)



SELECT ALL

- ☐ Play Therapy
- ☐ Individual
- ☐ Group
- ☐ Family
- ☐ Couple

Counselling CMS

CMSSessionType

Options:

1.	Expressive Therapy (Art, Body-based, Play)	SAVE	SET AS DEFAULT	↓	×	
2.	Counselling	SAVE	SET AS DEFAULT	↑	↓	×
3.	Individual	SAVE	SET AS DEFAULT	↑	↓	×
4.	Group	SAVE	SET AS DEFAULT	↑	↓	×
5.	Family	SAVE	SET AS DEFAULT	↑	↓	×

Add new value

ADD NEW VALUE

Counselling CMS



FOUNDED IN 1902

Children's Aid Society

- No need public appointment booking form



Haven Counselling Centre

Appointment Booking

1 Personal Particulars 2 Intake Information 3 Informed Consent

Personal Particulars (1/3) ENGLISH 中文

Name * Salutation *

Contacts * + ADD CONTACT

Nationality * Gender * ☐ Male ☐ Female

Date of Birth *

Marital Status * Religion *

Emergency Contacts * + ADD EMERGENCY CONTACT

← PREVIOUS NEXT →

Counselling CMS



FOUNDED IN 1902

Children's Aid Society

Manage CMS Intake Settings

36	Require Referral Source	Yes
37	Track Requested Counselling Mode	No
38	Require Requested Counselling Mode	No
39	Require Intake Clients	No
40	Require Referral Date	Yes
41	Require Referral Remarks	No
42	Enable Public Facing Appointment Form	No



Haven Counselling Centre

Manage CMS Intake Settings

36	Require Referral Source	Yes
37	Track Requested Counselling Mode	Yes
38	Require Requested Counselling Mode	Yes
39	Require Intake Clients	Yes
40	Require Referral Date	No
41	Require Referral Remarks	No
42	Enable Public Facing Appointment Form	Yes

Counselling CMS



FOUNDED IN 1902

Children's Aid Society

- No need Venue Rental Module



Haven Counselling Centre

- Need Venue Rental Module
- Different kinds of counsellors are charged different rates

Counselling CMS

New Venue

Name *

Harmony Room

Capacity

2

Default Rental Rate *

\$ 0

/hour

Rental Rates

+ ADD RENTAL RATE

Role

Tenant Counsellor



Rental Rate

\$ 30

/hour

Effective From

16/07/2025



DELETE

Role

Associate Counsellor



Rental Rate

\$ 20

/hour

Effective From

16/07/2025



DELETE

SUBMIT

RESET

CANCEL

Counselling CMS

New Venue Rental

 Bookings of the same venue must be at least 15 minutes apart when creating new booking.

Date *

29/07/2025



Start Time



End Time



Venue *



Today

July 29, 2025

	Harmony Room  2 \$ 0.00/h	Play Room  3 \$ 0.00/h	Synergy Room  2 \$ 0.00/h	Big Room  10 \$ 0.00/h
8am				
8:30am				
9am				
9:30am		Occupied 9:30 - 10:45 Play Room		
10am				
10:30am				

Counselling CMS

New Venue Rental

 Bookings of the same venue must be at least 15 minutes apart when creating new booking.

Date *
24/07/2025

Start Time
9:15 AM

End Time
10:15 AM (1h)

Venue *
Harmony Room

Venue Rental Rate
\$ 20.00 /hour

Total Rental Fee
\$ 20.00

< > Today

July 24, 2025

	Harmony Room 👤 2 \$ 20.00/h	Play Room 👤 4 \$ 17.00/h	Synergy Room 👤 2 \$ 14.00/h
8am			
8:30am			
9am			
9:30am	9:15 - 10:15		
10am			

Counselling CMS

MY TODOS ⁸	SUPERVISEE TODOS	ALL TODOS
-----------------------	------------------	-----------

Filters

Action Required

Search...

SEARCH

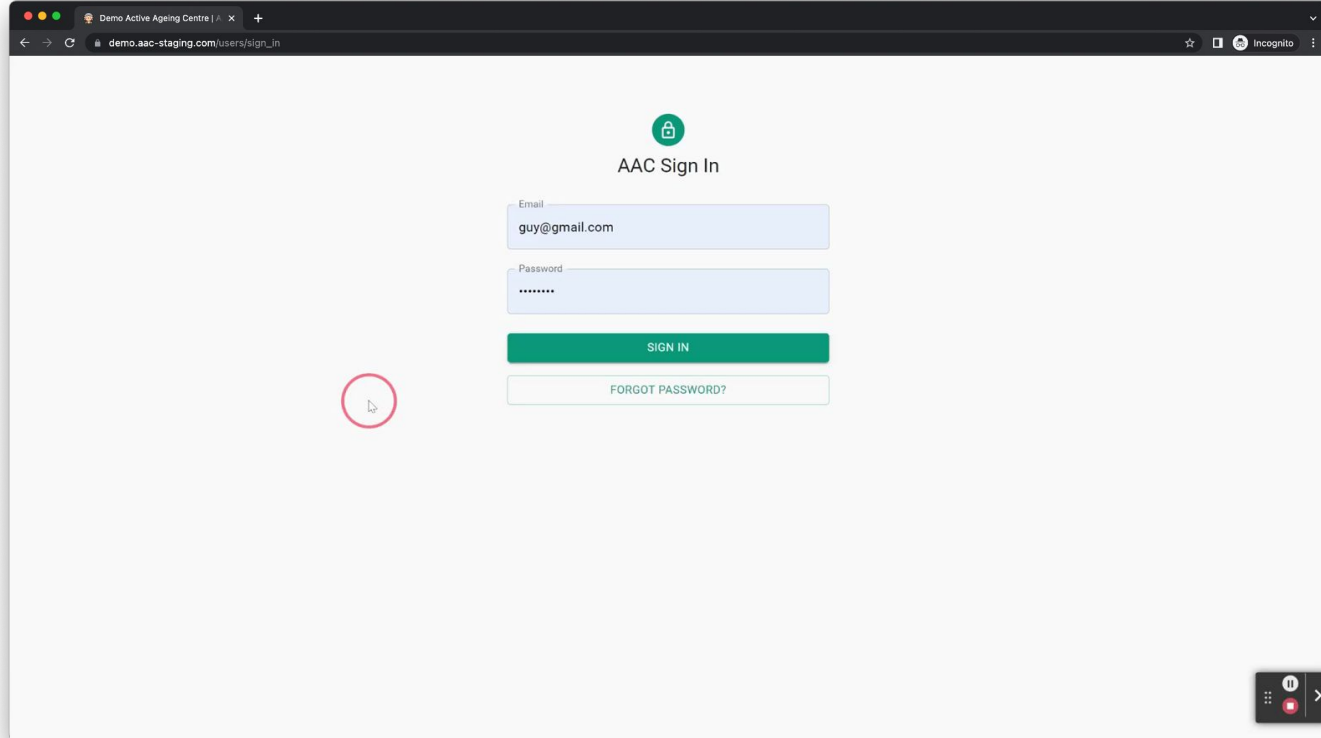
1-8 of 8

S/N	Action Required	Due Date	Action
1	Assign a case worker for Intake CAS/18/07/4-1 (Person Client 12)	Sat, 12 Jul 2025	>
2	Assign new staff for Case Transfer of Case ID12 (Person Client 12)	Tue, 15 Jul 2025	>
3	Approve closure request for Case ID19 (Person Client 17)	Wed, 16 Jul 2025	>
4	Promote Intake CAS/18/07/1-1 to a case (Person Client 11)	Fri, 18 Jul 2025	>
5	Accept / Reject assigned Case ID10 (Person Client 10)	Fri, 18 Jul 2025	>
6	Accept or Reject Intake CAS/18/07/2-1 (Person Client 12)	Mon, 21 Jul 2025	>
7	Approve transfer request for Case ID6 (Person Client 6)	Wed, 23 Jul 2025	>

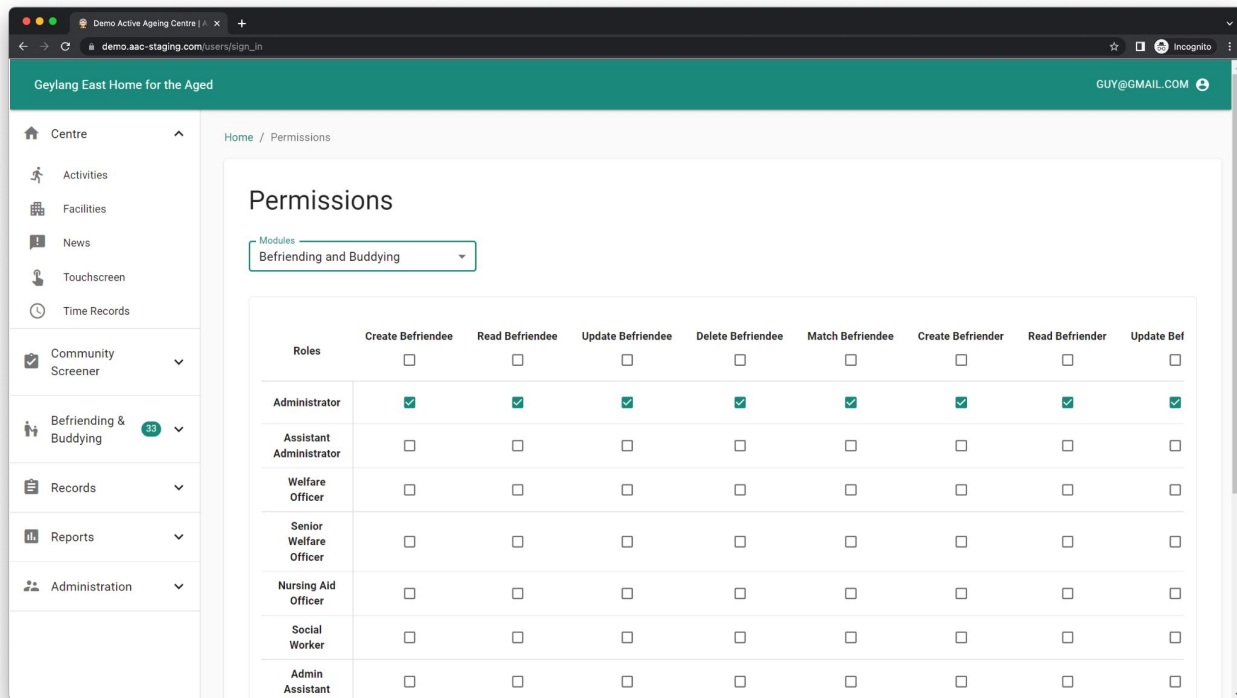
A photograph of two women smiling. The woman on the left is younger with short dark hair, wearing a dark red top. The woman on the right is older with short white hair, wearing a blue t-shirt. They are positioned in front of a blurred background that appears to be a painting or mural of a landscape with trees and a path.

General Features

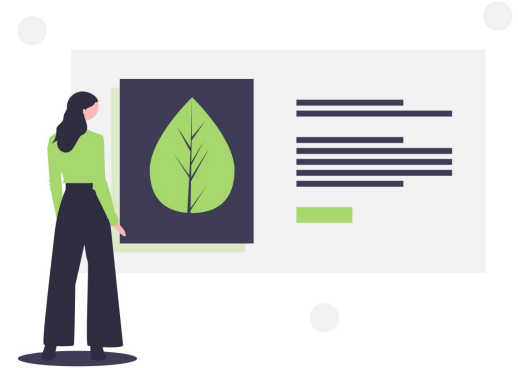
2FA



Role-Based Access Control



Expectations



Important Dates

- Internship starts 11th May 2026 (Monday after the end of exam week)
 - Must be physically in Singapore for the first 2 weeks of the internship to facilitate onboarding

Commitment Levels

- **5 days a week** during summer, **office hours (9am - 5pm)**
 - Members have worked overtime to finish their work
- Up to **10 days of leave**, cannot be taken in the first 2 weeks
- **Required** to be in Singapore
- **Cannot** be taken together with other full-time programmes such as Orbital and Special Term modules
- Can still participate in other programmes such as Orientation
- We fully support NS, so reservist is not an issue
- Orientation and reservist must be **declared during application** and **leave will be consumed**

Why CVWo



Why should you do CVWO?

- Pick up **web development** and **software engineering** skills
- Experience working in a **development team**
- Meet and develop applications for **actual clients**
- Contribute to the **community** with your programming abilities



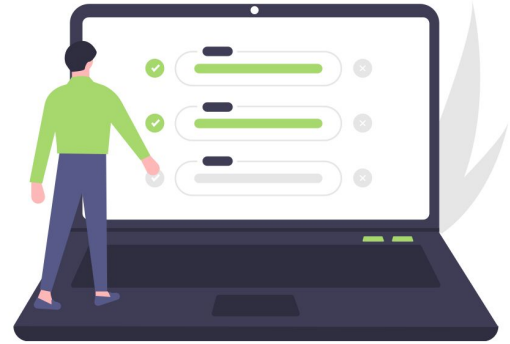
CVWO Benefits

- Provides **6 MCs** under CP3107 (*Graded)
- Satisfies either the **Industry Experience** or **Communities and Engagement (C&E) Pillar** requirement
- Stipend of **\$4800** kindly sponsored by GIC
 - \$2400 at the start, and \$2400 at the end

Looking Ahead

- Do well for CVWO in 2026 Summer
 - **Selected to be Team Lead** in 2027 Summer
- **Lead your own team** and learn:
 - Project Planning
 - Task Assignment
 - Leadership Styles
- Provides another **6 MCs** under CP3110
 - Also satisfies either the **Industry Experience** or **Communities and Engagement (C&E) Pillar** requirement

CVWO Application



Winter Assignment (Optional)

- Sent out in **early-December** & **due by late-Jan 2026**
- Involves building an application using Go, Typescript, React, PostgreSQL, and Git.
- **No prior experience is required** – it is meant to be a structured way to learn full stack development.
- Optional, but a good winter assignment will boost your application.

Application Details

- Opens: **19th January 2026** (Week 2 Monday)
- Deadline: **8th February 2026, 2359hrs** (Week 4 Sunday)
- Submit your resume, cover letter, and unofficial transcript
- Declare any events you are participating in over the summer
 - For example: Orientation, NS reservist, going back to home country, etc
- Finish HackerRank-styled take-home coding challenge (Emailed to you after 8 Feb)

Interview Details

- If shortlisted, we will send you an email with the details of the interview
- Interviews will take place over **Recess Week**
- Results should be released by the end of Recess Week

Warning: Cheating on Technical Assessment

- The technical assessment needs to be completed without external assistance, i.e. ChatGPT
- Some applicants were caught cheating this past year and were struck off.
- For this coming year, disciplinary action will be taken against such offenders because we consider this to be cheating.

Selection Criteria

- Good learning attitude is a must
- Winter assignment
 - Not a strict requirement, but a good winter assignment will most certainly help
- Grades for CS1101S, CS2030S, CS2103T will be considered, if you have taken them (its variants will be considered if you are not taking CS.)
- Prior experience
 - A plus, not a must
 - Describe in cover letter/resume

Bond Commitment

- The fellowship comes with a bond.
- This means that if you are selected to take on the role of Team Lead in Y2 Summer and you refuse, you will need to pay back the fellowship received.
- **This is necessary for the long term sustainability of the CVWO programme.**



<https://bit.ly/cvwo-mail-2026>

**For a more detailed overview of our
past projects, please visit**

<https://www.comp.nus.edu.sg/~vwo/projects.html>

**If you would like a copy of the slides,
please join our telegram channel**

https://t.me/CVWO_2526

If you have any queries, please email

cvwo.assignment@u.nus.edu

Q&A

Send your questions here:



forms.office.com/r/sCaGjt0E9D

Sign up for mailing list at:



bit.ly/cvwo-mail-2026